5. Devices

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5.1 Safe use of devices

Use of devices is an important sector of being safe online because phones, computers, tablets will have personal information, passwords, photos, sensitive information and financial details. If you lose a device or its stollen, infected with malware or connected to an unsecure device the information can then be damaged or accessed. This also involves keeping software updated, protecting devices with screen locks and encryption using software and being careful when downloading apps. Reports from the national cyber security centre state that having secure devices is an important part of protecting users and organisations from online threats. Use of safe devices essentially involves to be keeping access to devices protected using security software and encryption, making sure software is kept up to date also in general being careful your USB devices or public charging points. The National Institute of Standards and Technology explains that devices can face threats such as theft, malware and unauthorised access to stored information (Scarfone, 2007).

5.2 Keeping Your Software Updated

A software vulnerability is a weakness or problem in software that can potentially be used by an attacker, and a patch is updating software which is deigned to fix the vulnerability or other problem, exploiting can be a method or code that takes advantage of the vulnerabilities. It is very important to keep your software updated because cyberattacks can use known vulnerabilities to gain access to people's data or devices. People should enable automatic updates when you can an install security patches to reduce risk.

The NCSC explained that software updates help fix known security flaws and reduce the chances of attackers compromising devices. (NCSC, Managing-Deployed-Devices, 2021a).

Another important point is making sure devices are supported by their makers, older devices may potentially stop receiving security updates which means newfound vulnerabilities might not be fixed. This can leave users exposed to cyberattacks even if they regularly check for updates. The NCSC (NCSC, Managing-Deployed-Devices, 2021a) recommends replacing unsupported devices and software where it can replace these risks.

5.3 Encryption, Security Software, Screen locks

Security software like antivirus software helps to protect devices by blocking and being able to identify dangerous programs therefore antivirus software needs to be updated on the regular so it can be able to spot new pieces of malware. The NSCS recommend using antivirus software and other security software as part of a larger approach to device protection (NCSC, Managing-Deployed-Devices, 2021b).

For example, an Employee may leave their laptop unattended when taking a break, without a screen lock another person may access important documents or personal information, using a strong password or pin and locking the screen when leaving the device helps reduce the risk of unauthorised access (NCSC, Managing-Deployed-Devices, 2021e).

5.4 Risk of Installing data from Invalid Sources

This shows users not to straight up trust every app or website, downloading stuff from trusted developers or official stores, checking the permissions of the applications and making sure software is updated can help decrease the risks. The NCSC recommends controlling which third party applications are installed on devices to reduce potential threats. (NCSC, Managing-Deployed-Devices, 2021d)

5.5 Unknown USB Devices and Public Charging Points

Public charging port points can have security risks because some USB can transfer data and power, if a charging station has been modified to have malicious intention and can be potentially used to manipulate a connected device. Research by Spolaor at ACM (Spolaor, 2016) showed that USB charging connections could be exploited to take information from devices. To decrease the risk, you could instead use your own charging plug or carry a power bank.

Safe use of devices is important because it helps to protect your personal information and decreases the risk of cyberattacks, keeping software updated lessens attackers from manipulating vulnerabilities, while screen locks and encryption protect information if a device is stolen or lost. Security software helps to provide more protection against malware and downloading apps from valid sources decreases unnecessary risks. People using should be careful connecting unknown USB devices or using public charging points, by successfully following these precautions organisation and individuals improve device security and decrease the chances of information being infiltrated.